G AJAY KUMAR

Machine Learning Engineer
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SUMMARY

Fresher ML Engineer with a BSc Computer Science graduate. Built a Transformer encoder and end-to-end ML pipelines from scratch – achieving 87.9% accuracy and 0.954 AUC on the IMDB 50K benchmark. Proficient in TensorFlow, Scikit-learn, MLFlow and FastAPI; experienced in model training, evaluation, and REST API deployment.

TECHNICAL SKILLS

Programming: Python, SQL;
ML/DL Frameworks: Scikit-learn, PyTorch, TensorFlow, Keras, XGBoost, LightGBM
Deep Learning: Neural Networks, CNNs, RNNs, LSTMs, Transformer Architecture
NLP: NLTK, Spacy, Hugging Face Transformers, Text Classification, Transfer Learning, Fine-tuning
Data Analysis & Visualization: NumPy, Pandas, Matplotlib, Seaborn, EDA, Feature Engineering
MLOps & Deployment: MLFlow, FastAPI, Model Deployment, Data Pipeline, REST APIs
Database & Tools: MySQL, SQLAlchemy, Git, GitHub, VS Code, Jupyter Notebook

PROJECTS

End-to-End Diabetes Risk Prediction system | Python, SQL, Scikit-learn (Source Code)

- Built a modular, OOP-based end-to-end ML pipeline on the PIMA diabetes dataset – covering data preprocessing, feature engineering, stratified k-fold cross-validation, model selection, and REST API deployment using FastAPI and SQLAlchemy.
- Benchmarked Logistic Regression, Random Forest, Gradient Boosting, and SVM using stratified k-fold cross validation; selected Gradient Boosting based on AUC-ROC as the primary metric.
- Achieved 79% test accuracy with Gradient Boosting as the best-performing model and compared models using precision, recall, F1-score, AUC-ROC, and confusion matrix.
- Tracked all training experiments – hyperparameters, metrics, and model artifacts – using MLFlow.
- Designed a MySQL logging via SQL Alchemy ORM to persist in prediction logs enabling downstream analysis and audit trails.
- Diagnosed and resolved a data duplication bug in the MySQL prediction logging pipeline caused by repeated append-mode insert across API restarts – ensuring data integrity at scale.
- Deployed the Gradient Boosting model as a production REST API using FastAPI with pydantic request validation, supporting real-time single-record inference via JSON endpoints.

Transformer-Based Sentiment Analysis – From Scratch to API | TensorFlow, Keras (Source Code)

- Achieved **87.9% test accuracy and 0.954 AUC** on the IMDB 50K dataset by building a custom Transformer encoder in TensorFlow/Keras.
- Built a Transformer encoder from scratch implementing multi-head self-attention, sinusoidal positional encoding and stacked encoder layers with residual connections and layer normalization, following the original Attention is All You Need formulation.
- Engineered an end-to-end ML pipeline covering text preprocessing, custom tokenization (20k vocab), stratified train/val/test splitting, and model training with early stopping, learning rate scheduling, and Model Checkpoint callbacks.
- Tracked all hyperparameters, per-epoch metrics (accuracy, AUC, precision, recall), and model artifacts across experiments using MLFlow, enabling fully reproducible training runs.
- Deployed the trained model as a production REST API using Fast API, exposing /predict and /predict/batch endpoints with structured JSON responses including confidence scores and class probabilities.

EDUCATION

DR. BRR Govt. Degree College **Nov 2021 – July 2024**
Bachelor of Science in Computer Science
Coursework: Linear Algebra, Database Management Systems, Web Technologies.

CERTIFICATIONS

Python for Data Science, AI & Development - Coursera (IBM), 2025 (Link)
Data Analysis with Python - Coursera (IBM), 2025 (Link)
Machine Learning with Python - Coursera (IBM), 2025 (Link)
Introduction to Deep Learning & Neural networks - Coursera (IBM), 2025 (Link)